

Review

The clinical frailty scale as a predictor of orthopaedic outcomes: a narrative review

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ABSTRACT

Introduction: The Clinical Frailty Scale (CFS) is a 9-point scaling system used to categorise the frailty of patients. The CFS is well-established as a prognostic tool for decision-making within healthcare settings. However, the relationship between the CFS as a predictor for orthopaedic outcomes is limited. This review aims to provide an overview of the efficacy of the CFS as a prognostic tool for predicting orthopaedic outcomes.

Methods: Systematic review using PRISMA checklist (PROSPERO registered: CRD42023456648). Ovid and PubMed databases were searched using defined search terms to identify English language papers between 2007 and June 2023 which fit the inclusion criteria. Abstract screening was carried out independently and included studies proceeded to full-text review.

Results: 10 studies were identified. Studies used a range of outcome measures to assess success, including gross outcomes like mortality rates, as well as more specific functional outcomes, such as joint functionality scores. Studies identified that higher CFS scores correlate to poorer outcomes within orthopaedic patients. These include higher rates of mortality (41.7 % at one-year post proximal femur fracture for CFS ≥ 7), longer length of hospital stay and increased risk of adverse events post-procedure (both increased linearly from CFS 1 to 4). Additionally, the CFS was shown to be a strong prognostic tool when compared to other frailty scales. The number of studies that evaluated the relationship between the CFS and joint functionality scores is limited.

Conclusion: Higher CFS scores are associated with poorer orthopaedic outcomes. However, it is difficult to quantify the true impact due to the limited number of high-quality studies. Further work to characterise the relationship between both gross and functional outcomes associated with the utilisation of the CFS in orthopaedic settings is essential to ascertain the utility of this simple score to improve resource allocation and provide effective consent to patients.

Introduction

The Clinical Frailty Scale (CFS) is a scoring tool used by clinicians to quantify the degree of frailty in patients. Previous research has demonstrated a correlation between high CFS scores and poor patient outcomes, underpinning its relevance in providing care for the ageing population. [1] The CFS is a 9-point system scaled from 1 (very fit) to 9 (severe frailty with terminal illness) A person with a score of ≥ 5 is

considered frail. [2]

Frailty is a multisystem syndrome characterised by a reduction in physiological reserve and heightened susceptibility to poor health outcomes. The British Orthopaedic Association states that patients with a CFS of 5 or more should be managed using an established frailty pathway. [3] This includes a thorough geriatric assessment within 72 h of injury and multidisciplinary input into care. Previous work highlighted that frail patients have a 2.54 times higher chance of

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complications following surgery. [4] This study also stated that frailty positively correlates with hospital length of stay and patterns of discharge to assisted living facilities. The CFS is relied upon for its predictive success in the outcome of frail patients and to guide care decisions. A comprehensive literature review found the CFS to be predictive of outcomes in 74 % of cases of all frail, hospitalised patients; it had a strong association with mortality, complications, comorbidity, length of stay and cognition. [1] However, its utility in the context of orthogeriatric outcomes requires further clarification.

The objective of this review is to analyse the current literature focusing on the association between CFS and orthopaedic patient outcomes. This review will consider the benefits and disadvantages of the CFS as a predictor in orthopaedic surgery and will provide orthopaedic surgeons and perioperative clinicians insight into how they may use this tool to manage frail patients in their practice.

Methods

Review protocols

The following review protocols directed this literature review: the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement, [5] the PRISMA checklist, [6] and the Best Evidence Medical Education Guide No. 13 ‘Conducting a Best Evidence Systematic Review. [7] Ethical approval was not required for this study, as it was a review of published literature. This review was registered with the International Prospective Register of Systematic Reviews (PROSPERO) (CRD42023456648).

Data sources and search terms

We searched OVID and PubMed databases for literature sources. Our search terms for both engines were as follows:

((((((orthopaedic*) OR (orthopedic*)) OR (orthopedic surgery)) OR (orthopaedic surgery)) OR (orthopedic procedures[MeSH Terms])) OR (orthopaedic procedures[MeSH Terms])) AND (clinical frailty scale) between 2007 and June 2023. Our manual search covered the reference lists of potentially relevant papers and other frailty systematic reviews. We considered articles that included studies that described the outcomes of orthopaedic patients with a recorded CFS score.

Inclusion and exclusion criteria

We excluded articles which were not related to orthopaedic surgery, had no inclusion of the CFS and were non-English language (Table 1). Two authors (H.J. and I.H.) reviewed the abstracts and titles of papers found independently.

Articles that satisfied the predetermined criteria were chosen for

examination. With any discrepancy, the abstract was discussed further until a consensus was reached. In cases where there was a lack of agreement on the initial eligibility of abstracts, another author (A.P.) conducted an independent review to determine whether they fulfilled the criteria. Articles for full-text review were shared on a Microsoft Excel spreadsheet. Extracted data included a summary of the paper, methods, results, strengths and weaknesses.

Results

Pubmed yielded 124 potentially relevant articles. Medline OVID yielded 16 potentially relevant articles. Of which there were 15 overlapping studies. A further manual search found 1 further citation. A total of 10 suitable studies were included in this review following the screening of abstracts and full-text review. An accurate flowchart detailing our process can be found in Fig. 1.

The clinical frailty scale as a positive prognostic tool in orthopaedics

Multiple studies demonstrated the success of the CFS as a positive prognostic tool for predicting orthopaedic outcomes.

The studies utilised various outcome measures to evaluate the link between CFS and the outcomes of orthopaedic surgery, with mortality frequently being a key measure.

Narula et al. [8] aimed to assess the effectiveness of the CFS in predicting the likelihood of mortality within 30 days and one year following a proximal femur fracture (PFF) in elderly patients. The researchers conducted a retrospective analysis of fragility hip fractures between 2017 and 2018. A one-year follow-up was obtained for 502 cases. The mortality rates at 30 days and one year increased significantly with higher CFS scores ($p < 0.001$): 1.3 % for CFS 1 to 3, 4.4 % for CFS 4, 5.1 % for CFS 5, 7.1 % for CFS 6 and 14.1 % for CFS ≥ 7 mortality at 30 days. At 1 year the mortality rates were 3.8 % for CFS 1 to 3, 14.3 % for CFS 4, 23.9 % for CFS 5, 40 % for CFS 6 and 41.7 % for CFS ≥ 7 . Similar results were seen by Ikram et al. [9] The study included 1255 patients admitted to one institution and measured adverse outcomes in elderly patients with fragility hip fractures. The CFS scores were assigned by clinicians on admission, and outcomes such as 30-day survival were measured. The mortality risk within 30 days increased linearly for CFS points of 1 to 5 (Spearman’s Rho 0.12; $p < 0.001$) but plateaued for scores > 5 (rs(646) = 0.00; $p = 0.980$).

To gain a wider appreciation of CFS and its relationship to orthopaedic outcomes, rates of postoperative complication and length of stay were also often measured. Ikram et al. [9] indicated that postoperative complications (rs(605) = 0.21; $p < 0.001$) and length of stay (rs(589) = 0.31; $p < 0.001$) increased linearly for scores 1 to 4 but plateaued for scores > 4 . This study indicates that the CFS could offer valuable insights for less frail hip fracture patients yet might not accurately differentiate between higher CFS classifications.

Furthermore, Wang et al. [10] focussed on 87 patients awaiting elective joint replacement. This study defined 45.9 % of patients as robust (CFS 1–3), 49.4 % as prefrail (CFS 4) and 4.5 % as frail (CFS 5–8). Among the robust patients, 17.5 % had a hospital length of stay lasting more than 5 days, whereas, among frail patients, the percentage was higher at 48.9 %. The distribution of hospital length of stay was significantly different between the robust and prefrail-frail patient groups ($p = 0.005$). It was found that 10 % of patients classified as robust were moved to rehabilitation while a higher proportion, 27.7 %, of patients categorised as prefrail-frail required the same transfer ($p = 0.038$). Robust patients were found to be significantly correlated with a reduced hospital stay and a reduced number of discharges to the rehabilitation centre.

Complex modelling of data in a study by Chen et al. [11] determined that frail patients experienced more cumulative adverse events compared to robust patients. This prospective multicentre observational cohort design study investigated the relationship between frailty and

Table 1
Summary of inclusion/exclusion criteria used for this review.

Criteria	Inclusion	Exclusion
Participants	Studies involving patients managed by orthopaedic teams where outcomes of patients have been measured and CFS scores applied.	Studies where outcomes of orthopaedic patients were not measured. Studies without CFS application.
Study Type	Primary research that reports results suitable to the research question.	Studies that reported unsuitable outcomes unrelated to the research question.
Methodology	Quantitative, qualitative or mixed research reporting both gross and specific outcomes of patients.	Commentaries or article comments.
Timescale	Research published from 2007 onwards (changing of the 7-point CFS to 9-point CFS).	Research published before 2007.

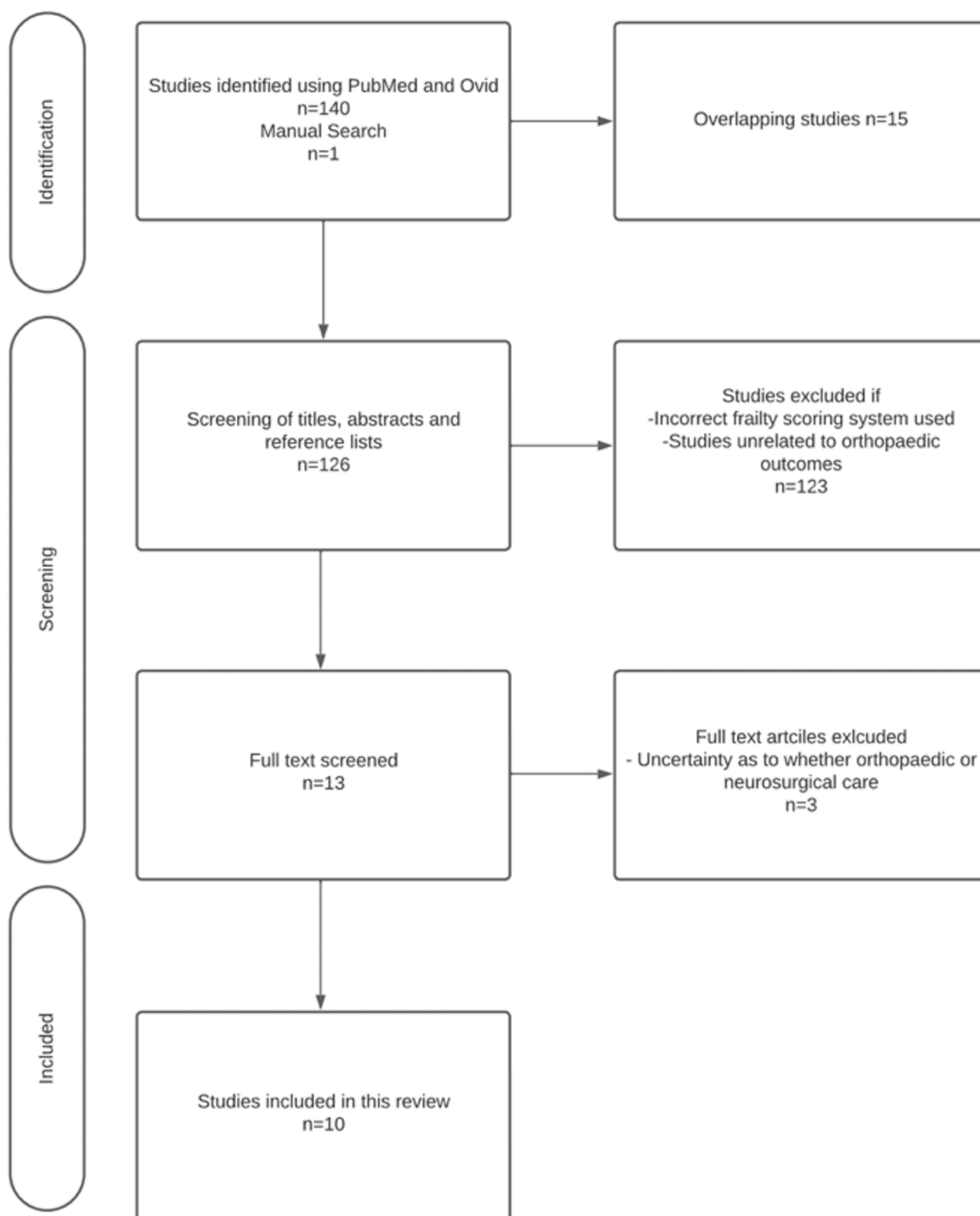


Fig. 1. A flow diagram, following the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA), illustrates the process of literature search and study selection.

emergency department visits, readmissions, and mortality at 1, 3, and 6 months postoperatively. The study included patients aged 50 years or older who received treatment for hip fractures, with a total of 245 patients selected. Frailty was assessed using CFS and categorised as robust (CFS 1–3), pre-frail (CFS 4–5), and frail (CFS 6–7). The data modelled

continuously linked the relationship between frailty and occurrences of mortality, initial readmissions, and visits to the emergency department within the 12 months following discharge.

Chan et al. [12] conducted a 5-year retrospective study with 423 hip fracture patients aged 65 and above at an academic trauma centre. They

examined discharge location, adverse events, and hospital stay length. The study revealed that CFS was a strong independent predictor for adverse discharge destination and complications, surpassing traditional risk factors like age, sex, time to surgery, and anaesthesia type. The study indicated a dose-response relationship between increasing frailty and hospital stay length (relative risk: 1.4, 1.6, and 2.2 for CFS 4, CFS 5, CFS 6–9, respectively; $P < 0.001$).

These gross outcome measures including mortality rates and hospitalisation duration, offer a comprehensive view of patient well-being and surgical efficacy. However, certain studies went beyond this to compare CFS against specific functional results stemming from orthopaedic procedures.

A retrospective cohort study by Moorthy et al. [13] compared 340 patients who underwent unilateral arthroscopic rotator cuff repair over two years. All patients received the same operation performed by the same shoulder surgeon. Functional outcomes were evaluated using various measurement tools such as the Oxford Shoulder Score (OSS), Constant Shoulder Score (CSS), University of California Los Angeles (UCLA) Shoulder Score, and visual analogue scale (VAS) for pain. These assessments were conducted before surgery and at 3, 6, 12, and 24-month intervals after the procedure. The CFS was significant only in predicting the 1-year CSS, UCLA Shoulder Score, and VAS score ($P < 0.05$).

A study by Jo et al. [14] aimed to assess whether total hip arthroplasty (THA) is a viable choice for appropriately selected elderly patients. The findings of this study demonstrate favourable outcomes and a relatively low rate of complications when THA is performed in patients aged ≥ 80 years with low CFS scores (< 5). A total of 82 patients aged ≥ 80 years underwent THA for displaced femoral neck fractures. THA was recommended for patients with a CFS score < 5 . Outcomes were evaluated using the modified Harris Hip Score (mHHS), VAS for pain, and patient satisfaction. Among the remaining 73 patients: the mean mHHS score preoperatively and one year postoperatively were 81.83 ± 12.87 and 80.57 ± 21.36 , respectively ($P = 0.67$); the VAS score was reduced six months postoperatively from 5.4 ± 1.3 , to 2.3 ± 0.9 ($P < 0.0001$); and 78.7 % of patients stated being very satisfied or satisfied with the results one year post-operatively.

The clinical frailty scale versus other prognostic tools

Furthermore, studies compared the CFS to multiple prognostic tools to predict orthopaedic outcomes. Narula et al. [8] compared the CFS to the American Society of Anaesthesiologists (ASA) and chronological age groups. The CFS showed a superior ability to predict mortality after PFF, with an area under the curve (AUC) of 0.699 (95 % CI 0.651 to 0.747), compared to the ASA classification (AUC 0.634; 95 % CI 0.576 to 0.691) and age groups (AUC 0.585; 95 % CI 0.523 to 0.648).

Wang et al. [10] compared the CFS to FRAIL Scale as screening instruments. As discussed, 45.9 % were classified as robust, 49.4 % as prefrail, and 4.5 % as frail with the CFS. In contrast, the FRAIL Scale categorised 13.7 % as robust, 62.0 % as prefrail, and 22.9 % as frail. Robustness, as determined by the CFS, was significantly associated with a shorter hospital stay ($p = 0.005$) and a lower likelihood of being discharged to a rehabilitation centre ($p = 0.038$). Proving superior to the FRAIL Scale which did not show significant associations with these outcomes.

A study by Wu et al. [15] aimed to determine which clinical assessment tools are most effective in predicting mortality risk following a hip fracture in chronic kidney disease (CKD) patients. A total of 397 patients met the inclusion criteria between June 2013 and December 2019. The CFS displayed the highest prognostic accuracy for both 30-day (AUC 0.91, 95 % CI 0.84–0.97) and 1-year mortality (AUC 0.93, 95 % CI 0.87–1.00) when compared to other prognostic tools such as the Charlson Comorbidity Index (CCI) and ASA.

Moorthy et al. [13] again compared CFS to other prognostic tools. The Modified Frailty Index (MFI) consistently emerged as a significant

predictor of functional outcomes at all postoperative time points up to 24 months ($P < 0.05$), whereas the CFS and CCI did not show the same level of significance. The MFI proved to be a superior tool compared to the CFS and CCI for predicting postoperative function in patients undergoing arthroscopic rotator cuff repair.

Thorne and Hodsgon et al. [16] aimed to report the outcomes of patients with hip fractures using data from two non-specialist hospitals in the UK. Data was collected from patients admitted with hip fractures over a 3-year period. In-patient mortality, 1-year mortality, and length of hospital were compared to the Nottingham Hip Fracture Score (NHFS) and the CFS. A total of 2422 patients were included, with 93 % undergoing surgery. The 30-day mortality rate was 5.8 % and the 1-year mortality rate was 23.5 %. The average hospital stay was 18.0 days. The NHFS had an AUC of 0.69 (95 % CI 0.64–0.74) for in-patient mortality, and the CFS had an AUC of 0.63 (95 % CI 0.57–0.69). For 1-year mortality, the AUC for the NHFS was 0.71 (95 % CI 0.68–0.73), and for the CFS, it was 0.67 (95 % CI 0.64–0.71). Neither score was able to predict extended hospital stay. Both the CFS and NHFS showed moderate discriminatory ability in predicting 1-year survival after a hip fracture. Similar results were seen in Ikram et al. [9] study where the CFS and NHFS had similar predictive values for 30-day mortality.

The clinical frailty scale and COVID-19

A recent study by Hall et al. [17] aimed to examine the effect of COVID-19 on frailty in patients with hip fractures. CFS scores were allocated upon admission and during follow-up. The outcomes measured included assessing the effects of COVID-19 on the duration of stay, post-discharge requirements, readmissions and discharge rates. The study included 68 COVID-positive patients and 141 COVID-negative patients. The examination demonstrated that COVID-19 was autonomously linked to heightened frailty, extended hospitalisation periods, elevated readmission rates, and a four-fold higher probability of patients not being able to return to their homes prior to the fracture. Additionally, the study showed that higher CFS scores on admission (OR 1.94, 95 % CI 1.46–2.67, $p < 0.001$) were independently associated with a failure to return home following the acute stay.

Discussion

This review demonstrates numerous studies have strongly advocated for the utilisation of the CFS as a predictive tool for assessing outcomes in orthopaedic surgery. Furthermore, studies have compared the CFS with other prognostic tools and consistently found it to be valid and in some cases superior. [8,15] However, due to the heterogeneity of outcome measures used in these studies, pooling data to characterise the true prognostic capabilities of the CFS in predicting orthopaedic outcomes becomes challenging. Outcome measures used can be broadly categorised into gross outcomes and specific functional outcomes of orthopaedic surgery summarised in Tables 2 and 3, respectively.

While multiple studies support the use of the CFS in predicting gross outcomes like mortality and hospital length of stay in orthopaedic surgery, the CFS ability to predict specific outcomes is limited. Our review identifies a literature gap in which further research is necessary to gain a comprehensive understanding of the relationship between the CFS and more specific functional outcome measures.

Frailty is a multi-system syndrome, and due to a lack of metabolic and musculoskeletal reserve frail patients are vulnerable to degenerative and traumatic bone and joint injury, meaning they make up a significant proportion of orthopaedic patients. The CFS is a good screening tool to broadly characterise a patient's preoperative functional status, but it is not an all-encompassing representation of frailty and does not measure all aspects of a patient's clinical frailty picture. As demonstrated by studies carried out on COVID patients we know that CFS can be influenced by factors such as acute infections, and perhaps if used in isolation can misrepresent a patient's true frailty status. Rather, it may best be

Table 2
Summary of studies and their gross outcome results.

Study	Outcome Measurement	Summary of result
Narula et al. [8]	Mortality at 30 days and 1 year of patients post PFF, length of stay, discharge to rehab, inpatient death	The 30-day mortality rate rose from 1.3 % in patients with a CFS score of 1 to 3 to 14.6 % in patients with a CFS score of 7 or higher. Similarly, the one-year mortality rate increased from 3.8 % in patients with a CFS score of 1 to 3 to 41.7 % in patients with a CFS score of 7 or higher.
Wang et al. [10]	Hospital length of stay and discharges to rehab centres	CFS had a significant association with hospital length of stay and discharge to a rehabilitation centre in patients undergoing total joint replacement.
Thorne and Hodgson et al. [16]	1 year mortality and hospital length of stay in hip fracture patients	Both the CFS and the NHFS demonstrate moderate and comparable ability to predict one-year survival. However, neither score was found to be effective in predicting length of hospital stay.
Wu et al. [15]	30 day and 1 year mortality in hip fractures	CFS predicted best accuracy for both 30 day and 1 year mortality
Chen et al. [11]	Adverse outcomes and mortality rates in hip fractures	Frail patients experienced a higher number of cumulative events for each adverse outcome compared to robust patients.
Chan et al. [12]	Discharge destination, in hospital complication and length of stay in hip fractures	Higher frailty scores were associated with adverse discharge destination, more hospital complications and longer hospital length of stay.
Ikram et al. [9]	30-day mortality, hospital complication and length of stay in hip fractures	Non-linear association found between CFS and outcomes measured. Results indicate difficulty in differentiating between higher CFS categories on outcomes.
Hall et al. [17]	Discharge destination and readmission	Higher CFS scores on admission were independently associated with a failure to return home following the acute stay.

Table 3
Summary of studies and their specific/functional outcome results.

Study	Outcome Measurement	Summary of result
Moorthy et al. [13]	OSS, CSS, UCLA Shoulder Score, and VAS.	CFS was significant only in predicting the 1-year CSS, UCLA Shoulder Score, and VAS score. However, the MFI was determined to be a more effective tool in predicting functionality compared to the CFS or CCI post-operatively.
Jo et al. [14]	The mHHS, VAS, and patient satisfaction.	THA with low CFS scores had favourable results and low rates of complications.

used as part of a wider comprehensive patient assessment. As research accumulates on the correlation between the CFS and postoperative functional status and mortality, the CFS could become a precise tool for tailoring personalised outcome measures. This could significantly refine the consent process, aligning it with established models like the NELA (National Emergency Laparotomy Audit) scoring system frequently used in general surgery to identify patients who need a greater degree of multidisciplinary input surrounding their perioperative care. [18]

The research predominantly concentrates on hip fractures, while some attention is given to shoulder surgery. Notably, upper limb surgery

consistently exhibits lower complication rates compared to lower limb procedures. [19] The heterogeneity in methods and limited studies of upper limb surgery make comparing studies difficult. This accentuates the necessity for additional investigation, particularly in light of the variability in mortality and hospital length-of-stay rates among different procedures, regardless of CFS scores, highlighting the need for joint-specific correlation data to the CFS.

The primary limitation of this study is the small sample size of the papers included for review. Whilst the CFS is widely used in acute inpatient hospital settings as a tool for clinicians, quantification of its relationship to orthopaedic surgery outcomes remains poorly recorded and evidence is limited based on the findings of this review. Another limitation is the methodology employed in some of the papers. The retrospective approach applied by several papers may not accurately capture the patient's frailty status pre-operatively. Furthermore, the grouping of CFS stages in some studies, such as combining stages 1–4, 5–6, and 7–9, can make it difficult to differentiate between different levels of frailty. Additionally, some studies scaled the CFS from 1 to 8 as seen by Wang et al. [10] and 1–7 by Chen et al. [11] introducing a lack of granularity to the results. This is particularly important as a study by Ikram et al. [9] reported a plateauing effect as the CFS increased, indicating that beyond a certain level of frailty, the impact on orthopaedic surgery outcomes may not be as pronounced. The extent of this effect and its implications need to be explored in further detail.

This review assesses mortality rates, hospital stay duration, functionality assessments, pain scores, and patient satisfaction as orthopaedic outcomes. Comparing the CFS with other prognostic tools would help to elicit the superior tool for predicting orthopaedic outcomes. Studies emphasise the practicality of CFS, easily applicable by a diverse healthcare team, reducing age-related bias in surgical decisions. Its broad applicability enhances feasibility and utility in orthopaedics. However, a limitation is the lack of injury/procedure-specific information. A prospective analysis focusing on specific injuries such as intertrochanteric fractures and their joint-specific function as well as gross outcomes related to preoperative CFS would be valuable, especially amidst growing healthcare demands. Tailoring orthopaedic care based on effective scoring systems for specific patient groups enhances decision-making and resource allocation efficiency. Research is warranted to establish the precise standard of care by the protocols set by the British Orthopaedic Association for the systematic management of frail patients within a dedicated frailty pathway. [3]

Current evidence has demonstrated that higher CFS scores are associated with poorer outcomes within orthopaedic patients. These include increased rates of mortality, length of hospital stay and risk of adverse events post-procedure. The CFS is a strong and often favourable prognostic tool when compared to other frailty scales. As research on the CFS advances, its potential impact on the consenting process and resource allocation for long-stay patients becomes increasingly evident. Enhanced insights into how CFS scores correlate with complications and recovery trajectories offer the opportunity for more precise consent discussions. Additionally, identifying patients at risk for extended hospital stays based on CFS scores enables efficient resource allocation, paving the way for targeted interventions and optimised care plans. Integrating CFS data into the consenting process facilitates informed decision-making, ensuring that frail patients and their families are well-prepared for potential challenges associated with orthopaedic surgery.

CRedit authorship contribution statement

Harri G. Jones: . **Isaac Hathaway:** Data curation, Formal analysis, Methodology, Writing – original draft. **Sean Glossop:** Methodology, Project administration, Visualization. **Hari Bhachoo:** Data curation, Formal analysis, Project administration. **Lucy Hoade:** Writing – original draft. **Joseph Froud:** Writing – original draft. **Lily Scourfield:** Supervision, Writing – review & editing. **Arwel T Poacher:** Conceptualization, Data curation, Formal analysis, Methodology, Supervision, Writing

– original draft, Writing – review & editing.

Declaration of competing interest

None declared.

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